

KOSZALIN, Poland

WMO: 121050

Lat: 54.2044N

Lon: 16.1553E

Elev: 35

StdP: 100.91

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.5	-8.8	-15.4	1.0	-10.7	-12.4	1.3	-7.8	11.2	5.8	10.1	5.5	2.2	150	0.544

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.8	28.1	19.3	26.1	18.7	24.3	18.1	20.8	26.0	19.7	24.1	18.8	22.8	3.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	13.9	22.7	18.1	13.0	21.7	17.2	12.3	20.8	60.3	26.3	56.4	24.1	53.4	23.0	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.9	7.7	6.6	-14.6	32.0	4.2	1.9	-17.7	33.4	-20.2	34.6	-22.5	35.6	-25.6	37.1		
			-15.0	22.9	4.2	1.3	-18.1	23.8	-20.5	24.5	-22.8	25.2	-25.9	26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	-0.1	0.8	3.3	8.1	12.1	15.3	17.7	17.9	14.2	9.5	4.9	1.4
	DBStd	7.38	4.55	4.25	3.81	4.20	3.97	3.26	3.05	2.89	2.94	3.36	3.42	4.34
	HDD10.0	1361	313	259	209	86	21	1	0	0	2	51	154	266
	HDD18.3	3563	571	492	466	309	198	103	49	44	129	274	403	524
	CDD10.0	923	0	0	2	28	86	159	240	244	127	34	2	0
	CDD18.3	83	0	0	0	1	6	12	30	30	4	0	0	0
	CDH23.3	604	0	0	0	12	55	111	227	182	19	0	0	0
	CDH26.7	136	0	0	0	1	8	30	59	36	2	0	0	0
Wind	WSAvg	3.1	3.6	3.6	3.4	3.2	2.9	3.0	2.8	2.7	2.8	3.0	3.3	3.6
Precipitation	PrecAvg	708	45	31	38	40	55	76	88	78	81	64	63	56
	PrecMax	1010	100	75	87	113	112	236	212	148	220	230	159	156
	PrecMin	483	2	5	7	5	9	21	13	9	9	13	15	6
	PrecStd	125	23	18	21	25	26	49	47	39	45	46	35	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.4	11.2	16.2	25.3	27.8	30.2	31.2	30.5	25.6	20.1	13.7	11.4
		MCWB	8.5	8.4	9.9	15.4	17.9	20.4	20.9	20.5	18.4	15.1	11.8	10.2
	2%	DB	8.2	9.2	13.0	20.5	24.8	26.4	28.3	27.1	23.1	17.6	11.8	9.7
		MCWB	6.9	7.2	8.5	13.1	16.8	19.0	19.9	19.0	16.9	14.2	10.1	8.4
	5%	DB	6.8	7.7	10.5	17.5	21.8	23.7	25.8	25.2	21.0	15.9	10.7	8.1
		MCWB	5.8	6.1	7.6	11.2	15.1	17.4	19.2	18.7	16.2	13.4	9.3	7.0
	10%	DB	5.5	6.3	8.6	15.1	19.3	21.3	23.7	23.4	19.0	14.3	9.5	6.8
		MCWB	4.5	5.0	6.3	10.2	14.1	16.4	18.4	18.1	15.1	12.4	8.3	5.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.9	9.2	11.2	15.8	19.1	21.8	22.4	21.8	19.4	16.1	12.4	10.5
		MCDB	10.2	10.6	14.4	24.0	25.1	28.7	28.1	27.2	23.5	18.6	13.4	11.1
	2%	WB	7.1	7.7	9.3	13.6	17.5	19.4	21.0	20.4	17.8	14.7	10.5	8.6
		MCDB	8.0	9.0	11.9	19.5	23.1	24.7	26.5	24.9	21.4	16.8	11.5	9.5
	5%	WB	5.8	6.3	7.8	12.0	16.0	18.2	19.9	19.5	16.6	13.6	9.4	7.1
		MCDB	6.7	7.4	10.4	16.5	20.9	22.6	24.4	23.5	19.6	15.4	10.5	7.9
	10%	WB	4.6	5.0	6.4	10.5	14.5	16.9	18.8	18.6	15.7	12.6	8.3	5.9
		MCDB	5.4	6.3	8.3	14.4	18.6	20.6	22.5	22.4	18.5	14.2	9.4	6.7
Mean Daily Temperature Range	5% DB	MDBR	3.8	4.7	6.2	8.2	8.6	8.0	7.8	8.0	7.3	6.0	4.1	3.7
		MCDBR	4.3	6.2	9.5	12.3	12.5	12.1	11.8	11.1	10.3	8.0	5.3	4.4
		MCWBR	3.9	4.8	6.2	6.6	6.5	6.3	5.5	5.4	5.3	5.1	4.2	4.1
	5% WB	MCDBR	4.2	5.5	8.5	10.9	11.3	10.9	10.3	9.4	8.7	7.2	5.0	4.3
		MCWBR	4.0	4.6	5.9	6.3	6.4	6.3	5.6	5.3	5.1	5.0	4.3	4.1
Clear-Sky Solar Irradiance	taub	0.312	0.338	0.365	0.403	0.401	0.396	0.420	0.418	0.378	0.361	0.340	0.309	
	taud	2.303	2.306	2.283	2.244	2.306	2.349	2.349	2.300	2.312	2.401	2.407	2.347	2.301
	Ebn at Noon	633	744	804	821	845	853	822	799	787	711	593	554	
	Edh at Noon	58	81	102	121	120	117	120	112	91	73	57	48	
All-Sky Solar Radiation	RadAvg	0.49	1.15	2.39	4.16	5.35	5.66	5.17	4.32	3.11	1.59	0.62	0.35	
	RadStd	0.07	0.17	0.25	0.50	0.60	0.42	0.60	0.39	0.33	0.23	0.09	0.04	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
Regional (35 neighbors)	+0.38	N/A	N/A	N/A	N/A	N/A	N/A	-130	+59	N/A	

Nomenclature: See separate page